

DIGITAL ELECTRONICS

ET 8301

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Lecture 8

REGISTER

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Introduction

- A **register** is a group of **memory cells** grouped together and considered a **single unit**. For example, an **8-bit register** could be used to store a **byte** of data.
- The register can be used to **simply store information for later use**, or the register can be designed to act on **the data as is the case of a shift register**.
- A shift register may **modify** the contents by **shifting data right or left**.
- The term **latch** may be used to describe the register used to store data.

Introduction

- Register they are commonly **constructed** using **flip-flops** (such as D flip-flops).
- A **buffer register** is a specific use of a storage device that holds data that are waiting to be transferred.
- For **example**, a buffer is used to **temporarily store** data while they are waiting to be used by a printer.
- A register has **no specified sequence** of states, except in certain very specialized applications.

Introduction

- A register, in general, is used **solely for storing** and **shifting data** (**1s** and **0s**) entered into it from an external source and typically possesses no characteristic internal sequence of states.
- A register is a digital circuit with **two basic functions**: **data storage** and **data movement**.
- A register is usually realized as **several flip-flops** with **common control signals** that control the movement of data to and from the register.
- An **n-bit register** is a group of **n flip-flops** and is capable of storing binary information containing **n bits**.

Introduction

- In addition to flip-flops, it may contain **combinational logic** which performs certain **data-processing tasks**.
- A register which is capable of **shifting its binary information** either **to the right or to the left** is called a **shift register**.
- The **storage capacity** of a register is the **total number of bits (1s and 0s)** of digital data it can retain. Each stage (flip-flop) in a shift register represents one bit of storage capacity.

Introduction

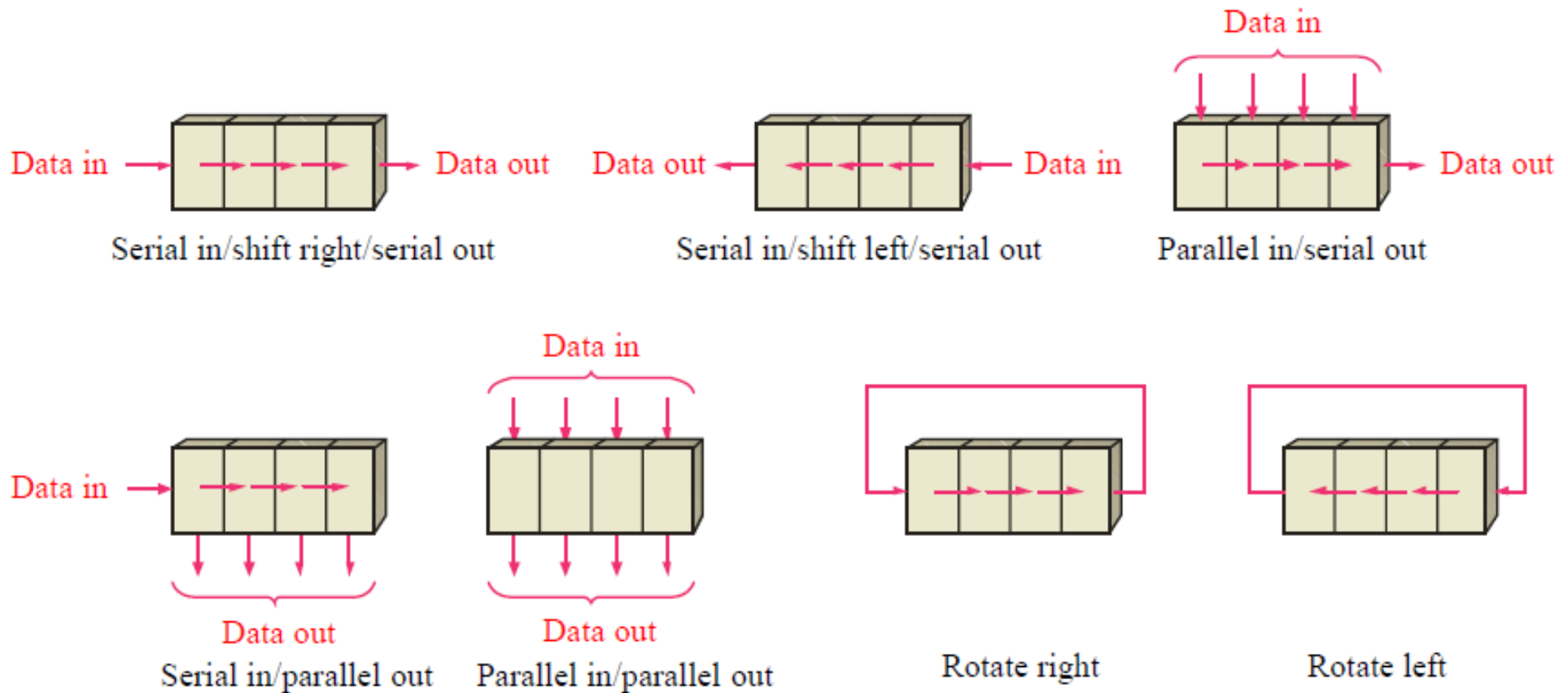
- The **shift capability** of a register permits the **movement of data** from **stage to stage** within the register or into or out of the register upon application of clock pulses.
- A shift register it consists of a **number of flip-flops cascaded together** with the **output of one flip-flop connected to the input of the next**.
- All flip-flops in the register receive a common clock pulse that causes the shift from one state to the next.
- A shift register is useful since a given binary number is **multiplied by 2** if its bits are shifted **one bit position** to the left and a **0** is inserted as the new least-significant bit.

Introduction

- Similarly, **a number is divided by 2**, if the bits are shifted **one position to the right**.
- A **typical example of a shift register** at work is found within a calculator. As you enter each digit on the keyboard, the numbers shift to the left on the display.

Introduction

➤ Some basic shift register **operations** are



Shift Registers

- Shift registers are constructed by **wiring flip-flops together**.
- We mentioned before that flip-flops have a **memory characteristic**. This memory characteristic is put to good use in a shift register.
- Instead of wiring shift registers by using individual gates or flip-flops, you can buy shift registers in integrated circuit (IC) form.
- In larger-scale digital devices (**microcontrollers, microprocessors**), the registers are **integrated into the chip design**.

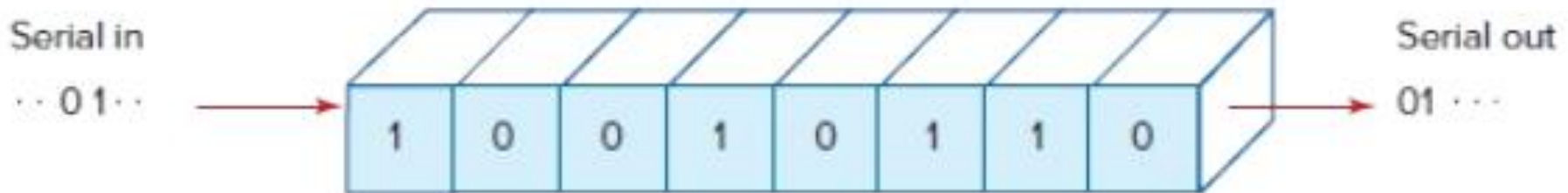
Shift Registers

- One method of describing shift register characteristics is by **how data are loaded into and read** from the storage units.
- Shift registers can be classified or categories based on four different types.
- These four types of shift registers are **based on data input and output (inputs/outputs)** which are : **serial in/serial out, serial in/parallel out, parallel in/serial out, and parallel in/parallel out.**

Shift Register

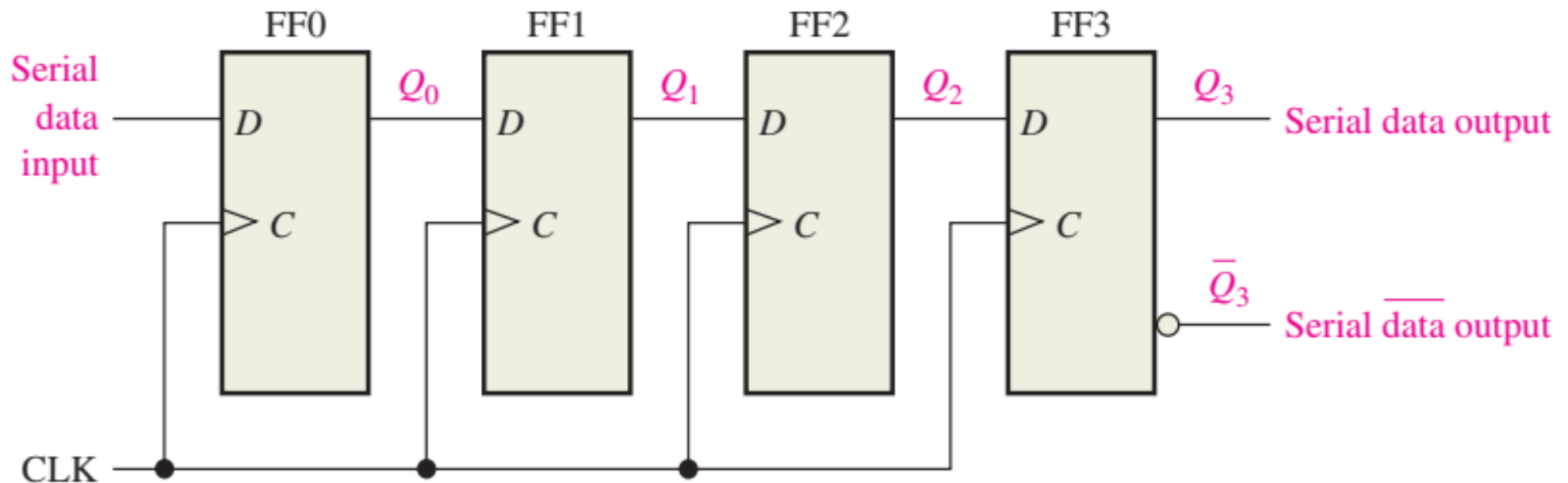
Serial In/Serial Out Shift Registers(SISO)

- This register **accepts data serially** that is, **one bit at a time** on a **single line**. It produces the stored information on its output also in **serial form**.
- For each clock pulse will move an input bit to the next flip-flop.



Shift Register

- Figure below shows a 4-bit device implemented with D flip-flops



Shift Register

- With four stages / flipflops, this register can store up to **four bits** of data.
- For **example** if we want to store the four bits **1010** into the register refer the figure above, beginning with the least significant bit.
- The register is initially clear. The **0** is put onto the data input line, making **D = 0** for **FF0**. When the **first clock pulse is applied**, **FF0** is **reset**, thus storing the **0**.

Shift Register

- Next the **second bit**, which is a **1**, is applied to the data input, making **D = 1** for **FF0** and **D = 0** for **FF1** because the D input of FF1 is **connected** to the **Q₀** output.
- When the **second clock pulse occurs**, the 1 on the data input is shifted into FF0, causing **FF0 to set**; and the 0 that was in **FF0 is shifted into FF1**.
- The **third bit**, a 0, is now put onto the data-input line, and a **clock pulse is applied**. The 0 is entered into FF0, the 1 stored in FF0 is shifted into FF1, and the 0 stored in FF1 is shifted into FF2.

Shift Register

- The **last bit**, a 1, is now applied to the data input, and a clock pulse is applied. This time the 1 is entered into FF0, the 0 stored in FF0 is shifted into FF1, the 1 stored in FF1 is shifted into FF2, and the 0 stored in FF2 is shifted into FF3.
- This completes the serial entry of the four bits into the shift register, where they can be stored for any length of time **as long as the flip-flops have dc power**.

Shift Register

➤ This can be illustrated in the table below

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	0	0	0	0
1	0	0	0	0
2	1	0	0	0
3	0	1	0	0
4	1	0	1	0

➤ If you want to get the **data out of the register**, the bits must be **shifted out serially to the Q_3 output**.

Shift Register

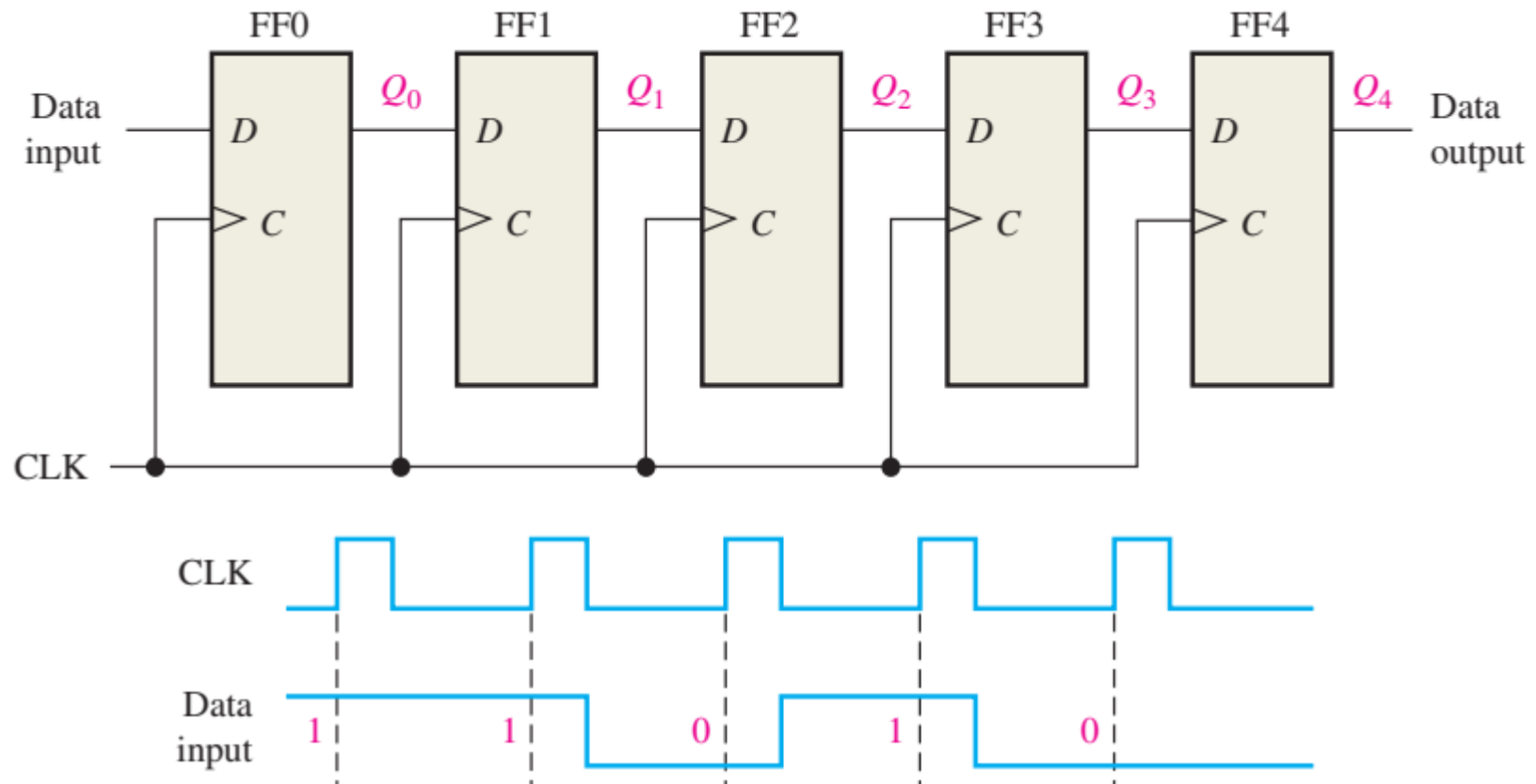
- After CLK4 in the data-entry operation just described, the LSB, 0, appears on the Q_3 output.
- When clock pulse CLK5 is applied, the second bit appears on the Q_3 output and so on.

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	1	0	1	0
5	0	1	0	1
6	0	0	1	0
7	0	0	0	1
8	0	0	0	0

➤ All :

Home work

- Show the states of the 5-bit register in figure below for the specified data input and clock waveforms. Assume that the register is initially cleared (all 0s).



Shift Register

- A **traditional logic block** symbol for an **8-bit serial in/serial out shift register** is shown in figure below



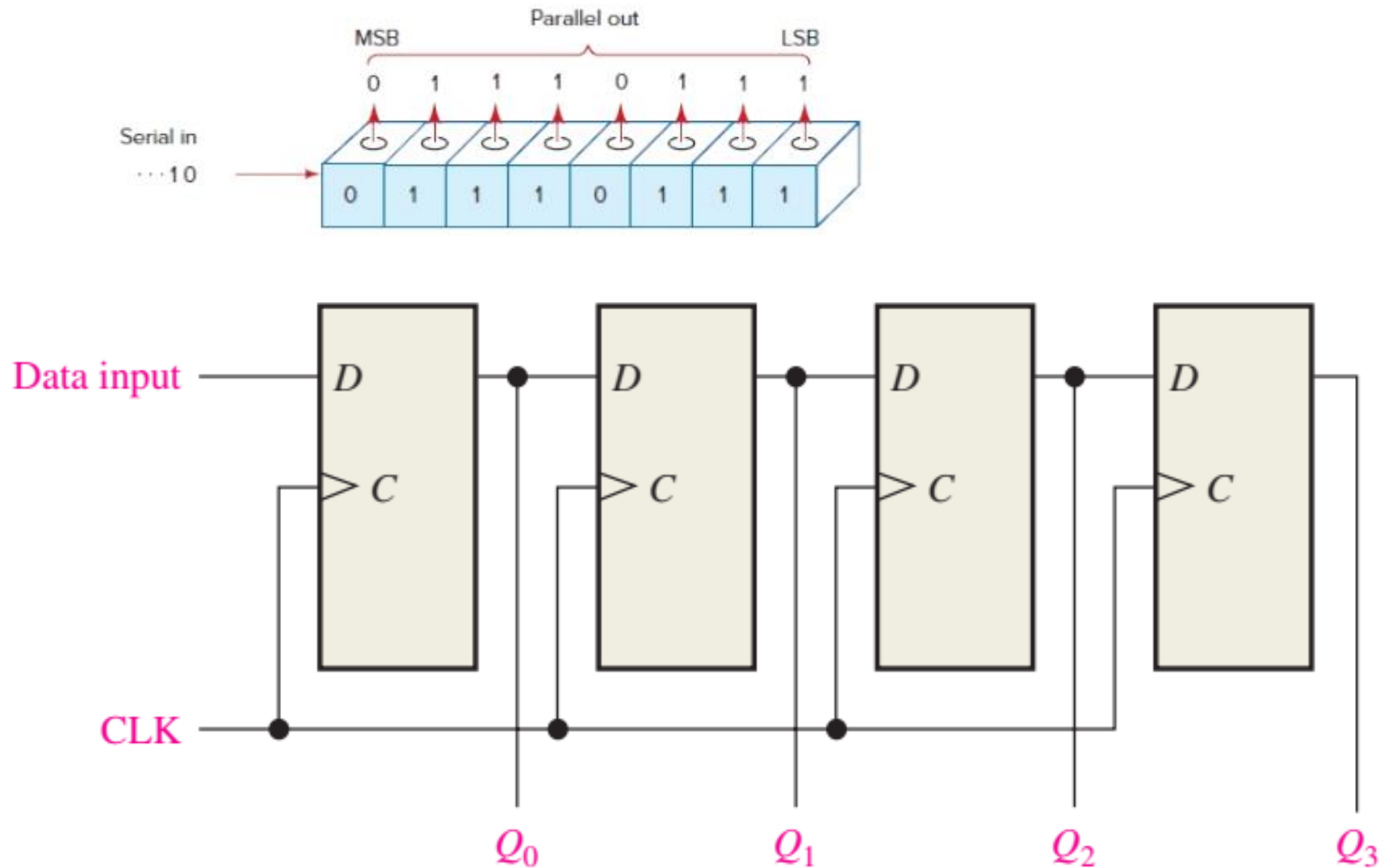
- The “**SRG 8**” designation indicates a **shift register (SRG)** with an **8-bit capacity**.

Shift Register

Serial In/Parallel Out Shift Registers(SIPO)

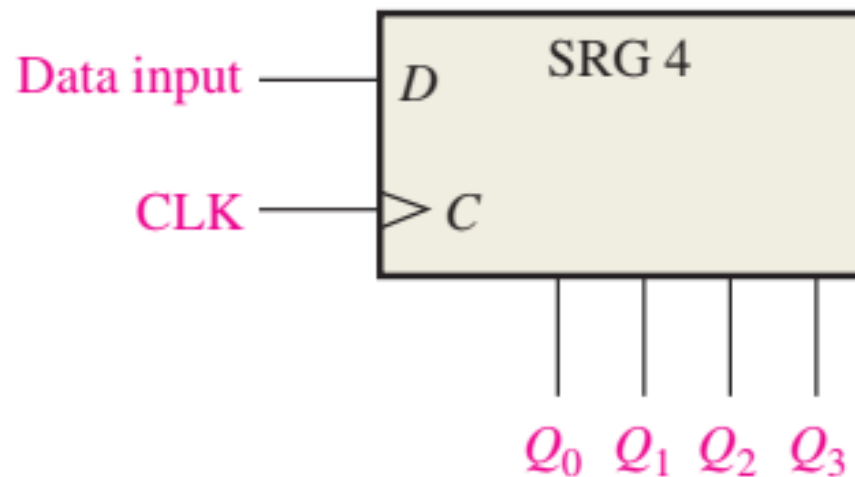
- For this register **data bits are entered serially** (least-significant bit first) into a serial in/parallel out shift register in the same manner as in serial in/serial out registers.
- The difference is the way in which the **data bits are taken out of the register**; in the parallel output register, the output of each stage is available.
- Once the data are stored, **each bit appears on its respective output line**.

Shift Register



Shift Register

- A **traditional logic block** symbol for an **4-bit serial in/parallel out shift register** is shown in figure below

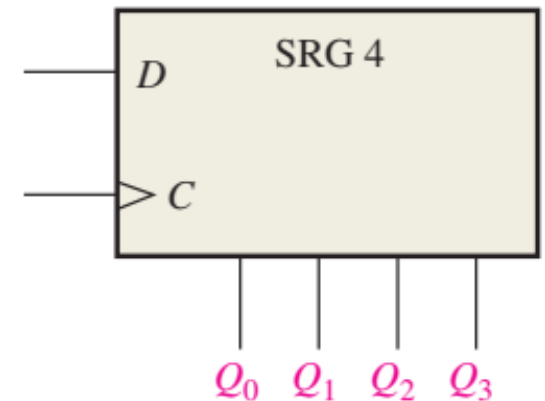
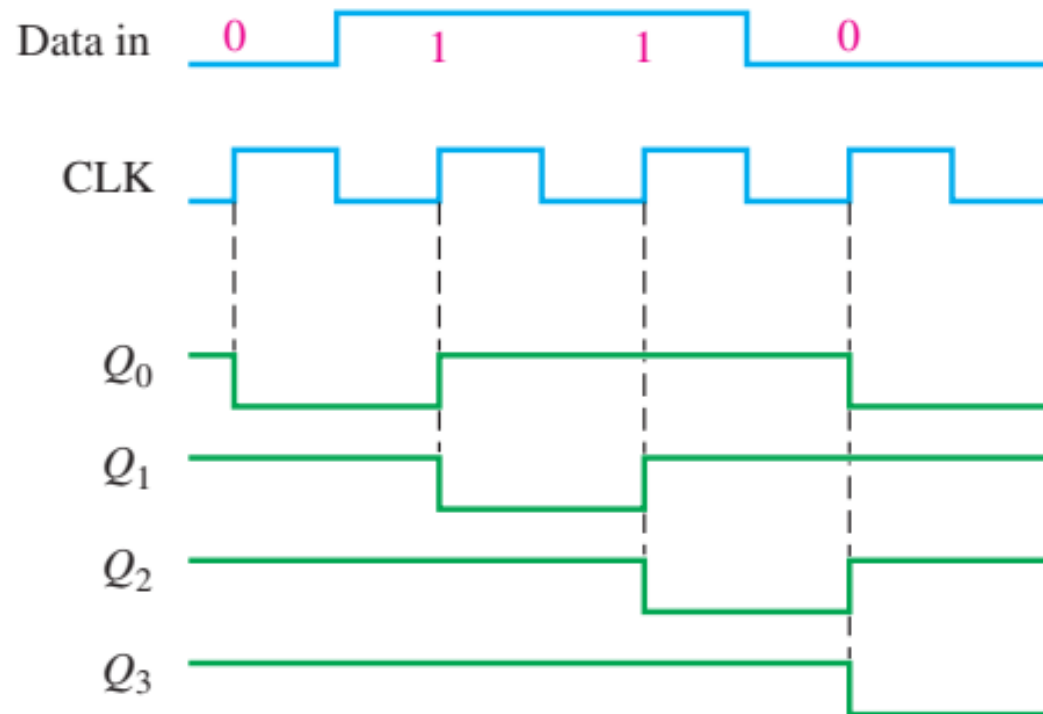


- The “**SRG 4**” designation indicates a **shift register (SRG)** with an **4-bit capacity**.

Shift Register

➤ Example

Show the states of the 4-bit register (SRG 4) for the data input and clock waveforms in Figure below the register initially contains all 1s.



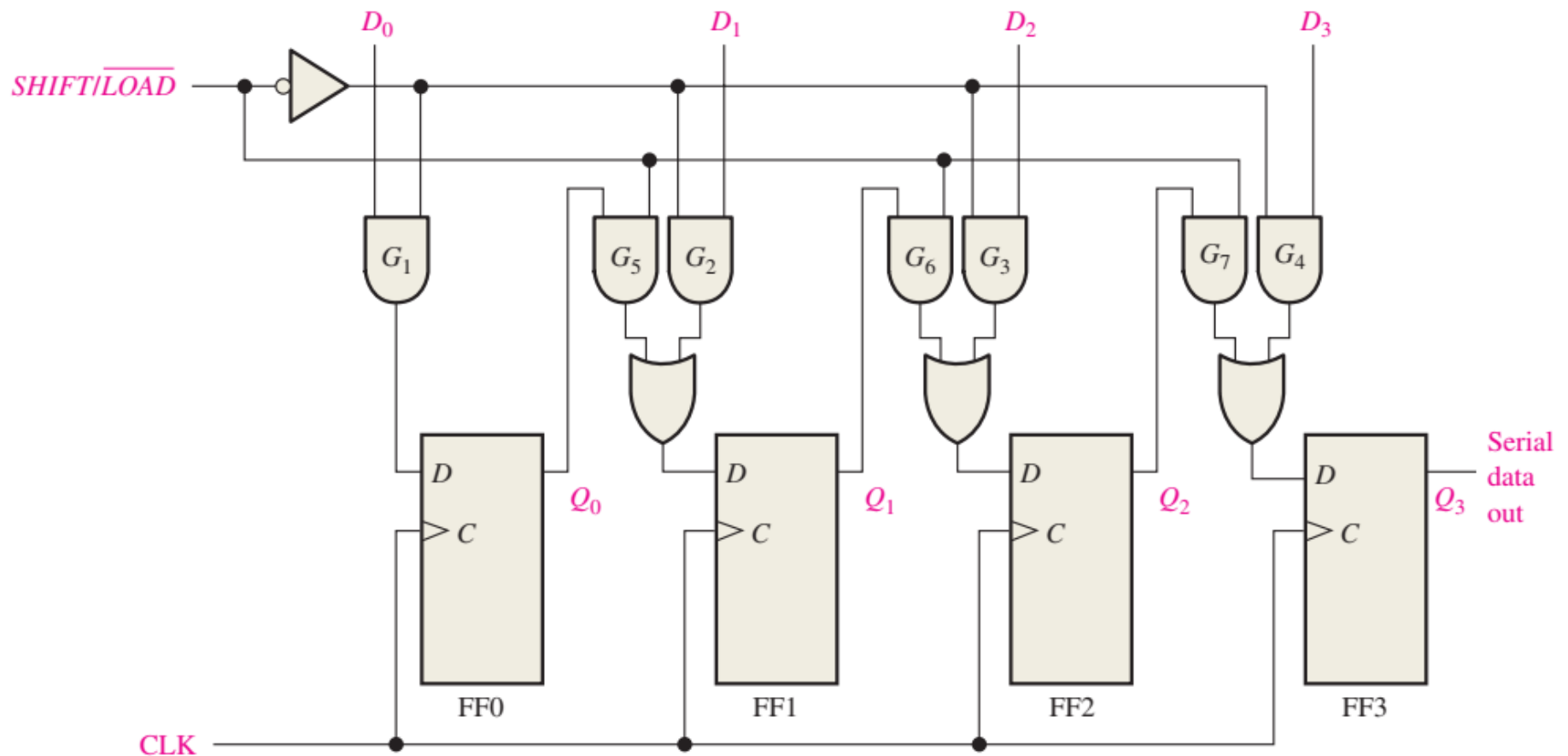
Shift Register

Parallel In/Serial Out Shift Registers(PISO)

- For a register with **parallel data inputs**, the bits are **entered simultaneously** into their respective stages on **parallel lines** rather than **on a bit-by-bit basis on one line** as with serial data inputs.
- The **serial output** is the same as in serial in/serial out shift registers, once the data are completely stored in the register.

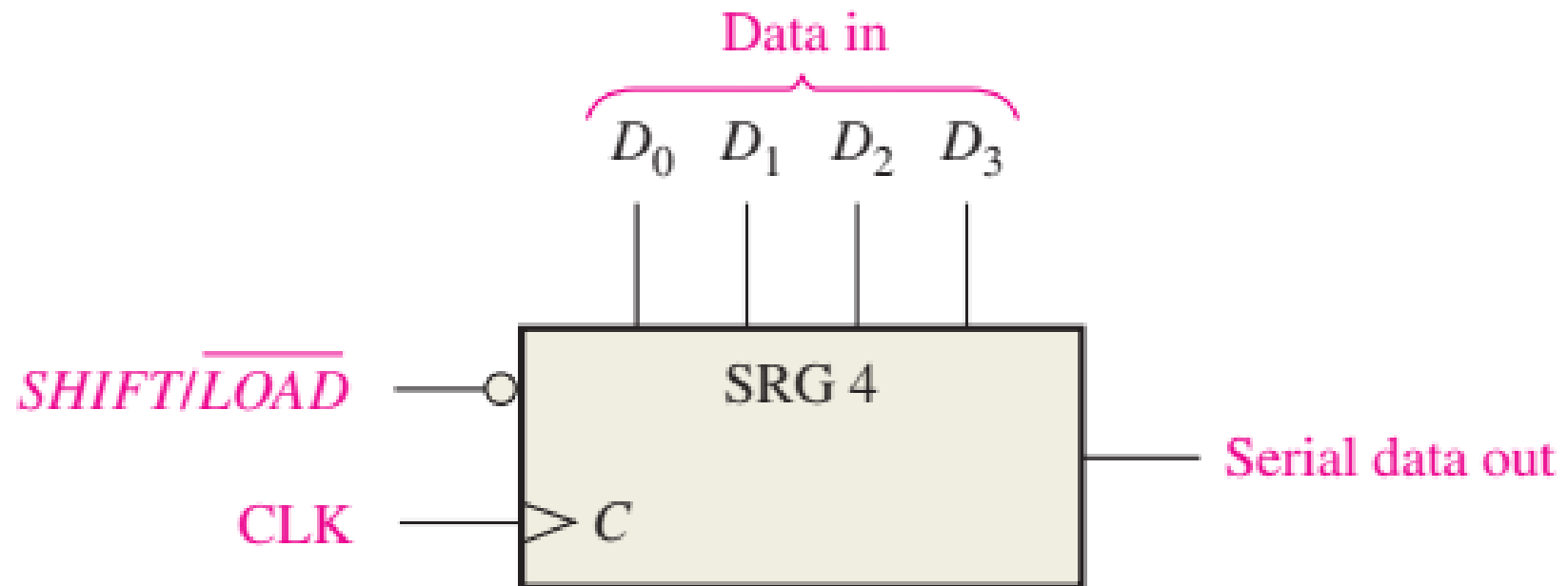
Shift Register

- Figure below illustrates logic diagram for a 4-bit parallel in/serial out shift register



Shift Register

- Figure below illustrates logic symbol for a 4-bit parallel in/serial out shift register



Shift Register

- There are **four data-input** lines, **D_0** , **D_1** , **D_2** , and **D_3** , and a **$\overline{\text{SHIFT/LOAD}}$** input, which allows four bits of data to **load** in parallel into the register.
- When **$\overline{\text{SHIFT/LOAD}}$** is **LOW**, gates **G_1** through **G_4** are **enabled**, allowing each data bit to be applied to the D input of its respective flip-flop.
- When a **clock pulse is applied**, the flip-flops with **$D = 1$** **will set** and those with **$D = 0$** **will reset**, thereby storing all four bits simultaneously.

Shift Register

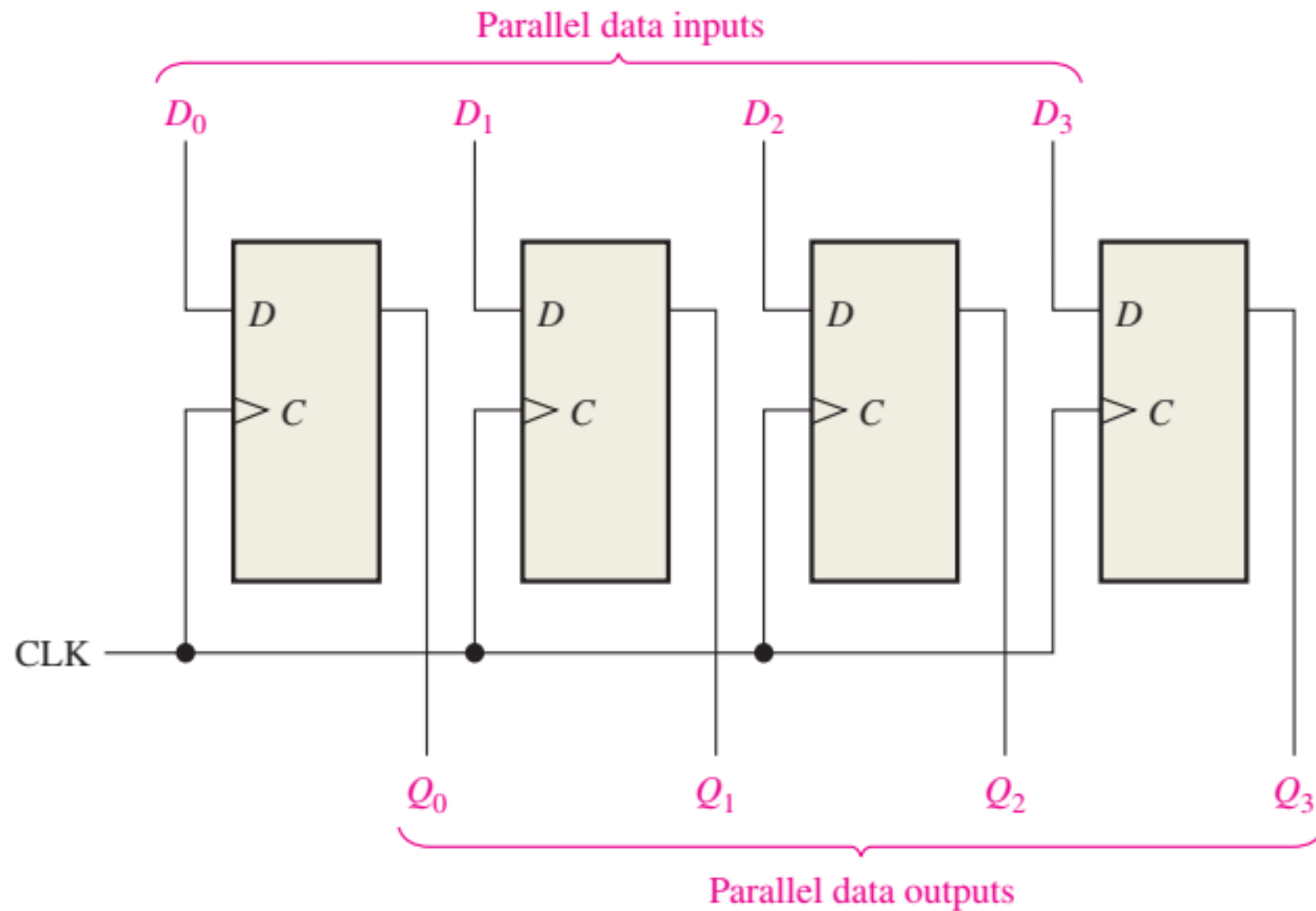
- When **SHIFT/ $\overline{LOAD}$** is **HIGH**, gates **G₁** through **G₄** are **disabled** and gates G₅ through G₇ are **enabled**, allowing the data bits to **shift right from one stage to the next**.
- The OR gates allow either the normal shifting operation or the parallel data-entry operation, depending on which AND gates are enabled by the level on the SHIFT/ $\overline{LOAD}$ input.
- **Notice** that FF0 has a single AND to disable the parallel input, D₀. It does not require an AND/OR arrangement because there is **no serial data in**.

Shift Register

Parallel In/Parallel Out Shift Registers(PIPO)

- The parallel in/parallel out register employs both methods. Immediately following the simultaneous entry of all data bits, the bits appear on the parallel outputs.
- The **PIPO** shift register takes the **data in and outputs it in parallel**.
- There is no shifting of data within the circuit, rather it simply acts as memory cells.

Shift Register

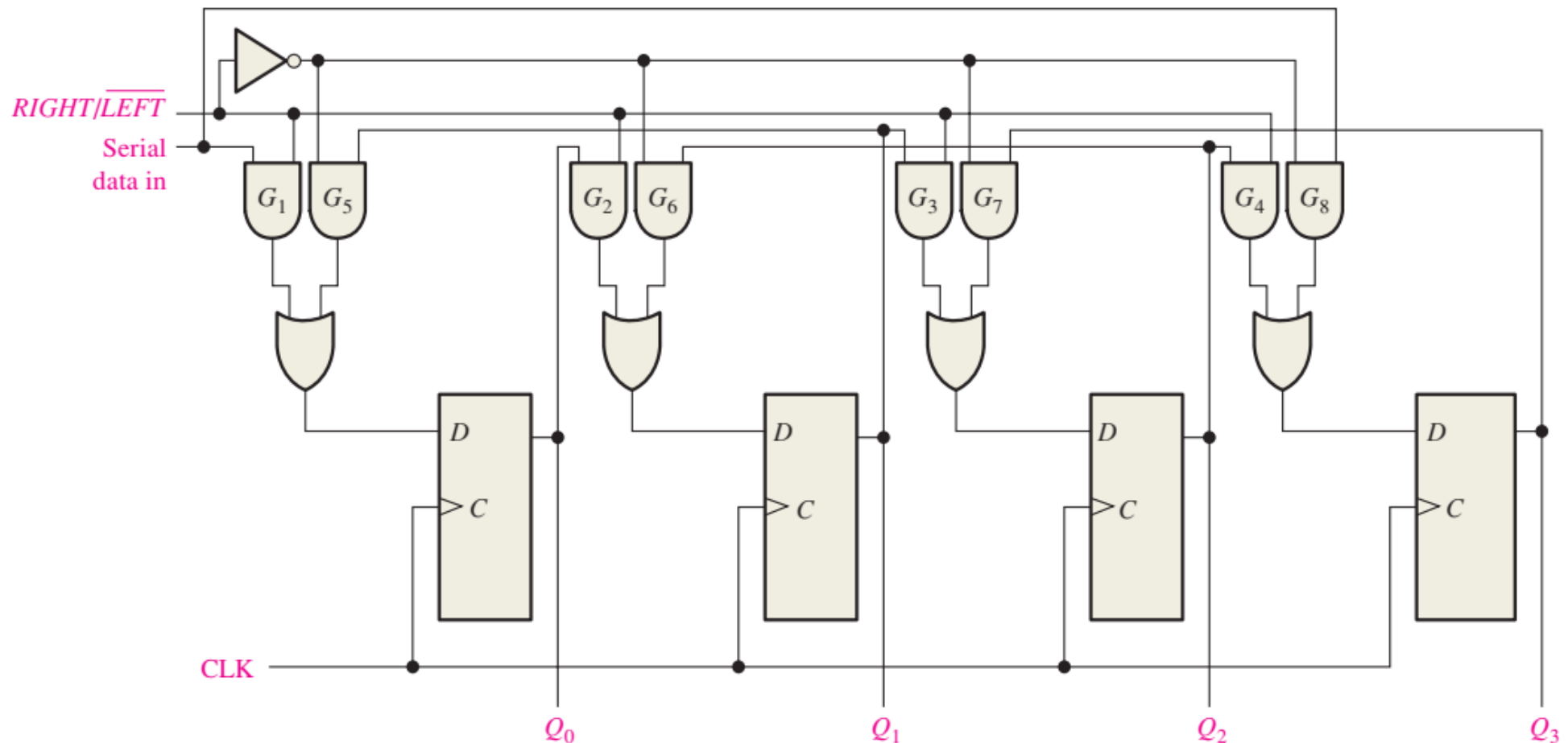


Bidirectional Shift Registers

- A **bidirectional shift** register is one in which the data can be shifted either left or right.
- It can be implemented by using **gating logic** that enables the transfer of a data bit from **one stage to the next stage to the right or to the left**, depending on the level of a control line.

Bidirectional Shift Registers

- 4-bit bidirectional shift register is shown in Figure below.



Bidirectional Shift Registers

- From the logic diagram above a **HIGH** on the **RIGHT/LEFT** control input allows data bits inside the register to be shifted to the right, and a **LOW** enables data bits inside the register to be shifted to the left.
- When the **RIGHT/LEFT** control input is HIGH, gates G_1 through G_4 are **enabled**, and the state of the Q output of each flip-flop is passed through to the **D input** of the following flip-flop.
- When a clock pulse occurs, the data bits are shifted one place to the **right**.

Bidirectional Shift Registers

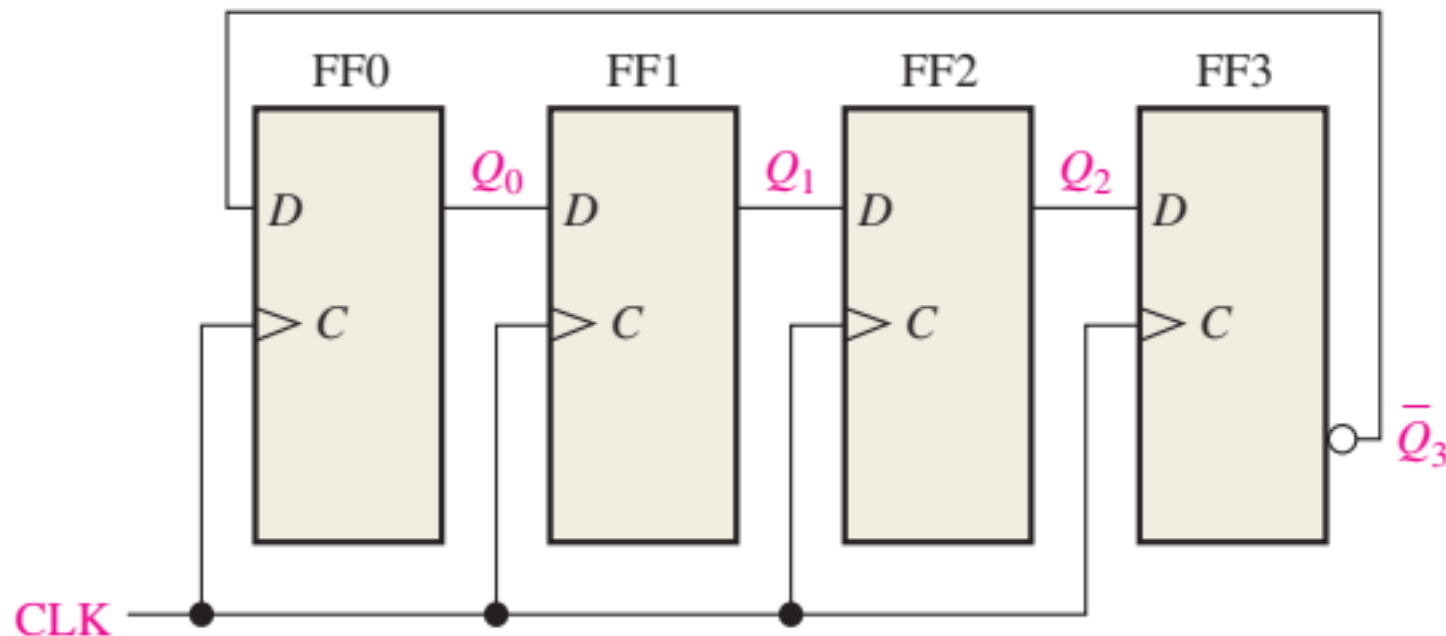
- When the **RIGHT/LEFT** control input is LOW, gates G_5 through G_8 are **enabled**, and the Q output of each flip-flop is passed through to the D input of the preceding flip-flop.
- When a clock pulse occurs, the data bits are then shifted one place to the **left**.

Applications of shift registers

- Shift registers are found in many types of applications.
- A **shift register counter** is basically a shift register with the **serial output connected back to the serial input to produce special sequences**.
- These devices are often classified as **counters** because they **exhibit a specified sequence of states**.
- Two of the most common **types of shift register counters**, the **Johnson counter** and the **ring counter**.

The Johnson Counter

- In a **Johnson counter** the **complement of the output of the last flip-flop** is connected back **to the D input of the first flip-flop** (it can be implemented with other types of flip-flops as well).



The Johnson Counter

- If the counter starts at 0, this feedback arrangement for a 4-bit device produces a characteristic sequence of states, as shown.

Four-bit Johnson sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1



The Johnson Counter

- You can notice that the **4-bit sequence** has a **total of eight states**, or bit patterns, and that **the 5-bit** sequence has a total of **ten states**.
- In general, a Johnson counter will produce a **modulus of $2n$** , where **n** is the **number of stages** in the counter.

The Ring Counter

- A **ring counter** utilizes **one flip-flop for each state** in its sequence.
- It has the advantage that decoding gates are not required. In the case of a 10-bit ring counter, there is a unique output for each decimal digit.

Homework

- Read chapter 8 of Digital Fundamentals by Floyd and do Problems found at the end of the chapter
- Read more about the application of register.

Home work

- What is a shift register, and why can such a circuit be used to multiply a binary number by 2^n .
- How many flip-flops are required to construct mod-12 ring and Johnson counters?
- What do the terms transparent; level triggered and edge triggered mean?



Next Lecture

COUNTER